CPE 381 Computer Graphics

Lab 5
3D Transformation

Dr. Taweechai Nuntawisuttiwong

Lab Setup

- Lab 5 3D Transformation
  - shaders
    - fragment.glsl
    - vertex.glsl
  - index.html
  - lab5.js
  - matrix.js

WebGL Cube Geometry

- Define 8 vertices of a cube (line 49 - 59)
- Use indexed drawing with **gl.TRIANGLES** (line 101)
- Add cube edges using **gl.LINES** (line 103)

Transformation Controls

- Translate
  - **w**: move up
  - **s**: move down
  - **a**: move left
  - **d**: move right
  - **r**: move forward (into screen)
  - **f**: move backward (out of screen)
- Rotate
  - **Arrow Up**: rotate around X-axis (+)
  - **Arrow Down**: rotate around X-axis (-)
  - **Arrow Left**: rotate around Y-axis (-)
  - **Arrow Right**: rotate around Y-axis (+)
  - **q**: rotate around Z-axis (-)
  - **e**: rotate around Z-axis (+)
- Scale
  - **t**: enlarge
  - **g**: shrink

matrix.js Functions

- matIdentity()
- matTranslate(tx, ty, tz)
- matRotateX(angle), mat4RotateY(angle), mat4RotateZ(angle)
- matScale(sx, sy, sz)
- matMultiply(a, b)
- matPerspective(fov, aspect, near, far)

Task

1. Implement transformation functions in gl-matrix.js
2. Render cube with transformation (line 95)
3. Add keyboard controls to modify transform values (line 111 - 124)

Deliverables

- Working WebGL application with interactive 3D transformation
- Manual implementation of matrix operations
- Responsive keyboard controls